

*Non-Calculator***Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.****You have 30 minutes.**_____
Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[\sqrt{6}(\cos 165^\circ + i \sin 165^\circ) \right] \left[\sqrt{2}(\cos 255^\circ + i \sin 255^\circ) \right]$

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

2. $\frac{-\sqrt{3} + i}{\sqrt{2} - i\sqrt{6}}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3. $(-1 - i)^3$

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

4. $x^3 + 8i = 0$

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

5. a) $z = i$	YES / NO	b) $z = 1 + i$	YES / NO

ANSWER KEY – Non-Calculator

SHOW ALL WORK FOR FULL CREDIT!

Every line shown. Point splits are an estimate of partial credit (halves are possible).

Find the product. Write the product in rectangular form, using exact values. (3 points)

Problem: $\left[\sqrt{6}(\cos 165^\circ + i \sin 165^\circ) \right] \left[\sqrt{2}(\cos 255^\circ + i \sin 255^\circ) \right]$

1. Multiply the r 's, add the θ 's.

$$r = \sqrt{6} \cdot \sqrt{2} = \sqrt{12} = 2\sqrt{3} \quad \theta = 165^\circ + 255^\circ = 420^\circ - 360^\circ = 60^\circ$$

$$2\sqrt{3}(\cos 60^\circ + i \sin 60^\circ) = 2\sqrt{3}\left(\frac{1}{2}\right) + 2\sqrt{3}i\left(\frac{\sqrt{3}}{2}\right) = \boxed{\sqrt{3} + 3i}$$

Trap: 165° and 255° aren't on the unit circle, but the sum is. Simplify $\sqrt{12} = 2\sqrt{3}$, and $\sqrt{3} \cdot \sqrt{3} = 3$.

1 pt r, 1 pt reduced angle, 1 pt exact rectangular.

Convert to trigonometric form, then divide. Write the quotient in rectangular form, using exact values. (3 points)

Problem: $\frac{-\sqrt{3} + i}{\sqrt{2} - i\sqrt{6}}$

2. Convert the top and the bottom separately (sketch each for the quadrant).

Top: $r = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{4} = 2 \quad \tan \theta = \frac{1}{-\sqrt{3}} \Rightarrow \text{ref } 30^\circ, \text{ QII} \Rightarrow 150^\circ$

Bottom: $r = \sqrt{(\sqrt{2})^2 + (-\sqrt{6})^2} = \sqrt{8} = 2\sqrt{2} \quad \tan \theta = \frac{-\sqrt{6}}{\sqrt{2}} = -\sqrt{3} \Rightarrow \text{ref } 60^\circ, \text{ QIV} \Rightarrow 300^\circ$

$$\begin{aligned} \frac{2 \text{ cis } 150^\circ}{2\sqrt{2} \text{ cis } 300^\circ} &= \frac{1}{\sqrt{2}} \text{ cis}(150^\circ - 300^\circ) = \frac{\sqrt{2}}{2} \text{ cis}(-150^\circ) \xrightarrow{+360^\circ} \frac{\sqrt{2}}{2} \text{ cis } 210^\circ \\ &= \frac{\sqrt{2}}{2} \left(-\frac{\sqrt{3}}{2}\right) + \frac{\sqrt{2}}{2} i \left(-\frac{1}{2}\right) = \boxed{-\frac{\sqrt{6}}{4} - \frac{\sqrt{2}}{4} i} \end{aligned}$$

Trap: rationalize $\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$. Negative angle: add 360°.

Find the power. Write the answer in rectangular form, using exact values. (3 points)

Problem: $(-1 - i)^3$

3. Convert, then De Moivre: $[r \text{ cis } \theta]^n = r^n \text{ cis}(n\theta)$.

$$r = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2} \quad \tan \theta = \frac{-1}{-1} = 1, \text{ QIII} \Rightarrow 225^\circ$$

$$\begin{aligned} (\sqrt{2} \text{ cis } 225^\circ)^3 &= (\sqrt{2})^3 \text{ cis } 3(225^\circ) = 2\sqrt{2} \text{ cis } 675^\circ \xrightarrow{-360^\circ} 2\sqrt{2} \text{ cis } 315^\circ \\ &= 2\sqrt{2} \left(\frac{\sqrt{2}}{2}\right) + 2\sqrt{2} i \left(-\frac{\sqrt{2}}{2}\right) = \boxed{2 - 2i} \end{aligned}$$

Trap: $(\sqrt{2})^3 = \sqrt{2} \cdot \sqrt{2} \cdot \sqrt{2} = 2\sqrt{2}$. Both parts negative is QIII (225°), not the 45° that $\tan^{-1}$ gives.

Find all complex number solutions of the equation. Write answers in rectangular form. (3 points)

Problem: $x^3 + 8i = 0$

4. Isolate x^3 first, then take cube roots.

$$x^3 = -8i = 0 - 8i \text{ (negative imaginary axis): } r = 8, \theta = 270^\circ$$

$$x = (8 \operatorname{cis} 270^\circ)^{1/3} = \sqrt[3]{8} \operatorname{cis} \frac{270^\circ}{3} = 2 \operatorname{cis} 90^\circ + 120^\circ$$

$$2 \operatorname{cis} 90^\circ = 2i \quad 2 \operatorname{cis} 210^\circ = 2 \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = -\sqrt{3} - i \quad 2 \operatorname{cis} 330^\circ = \sqrt{3} - i$$

$$x = 2i, -\sqrt{3} - i, \sqrt{3} - i$$

Trap: moving $+8i$ across makes it $-8i$, which is 270° , not 90° . Check: $(2i)^3 = 8i^3 = -8i$.

The fractal called the **Mandelbrot set**: to determine if a complex number $z = a + bi$ is in this set, repeatedly compute $z, z^2 + z, (z^2 + z)^2 + z, [(z^2 + z)^2 + z]^2 + z, \dots$. The complex number z does **not** belong to the Mandelbrot set if any of the resulting absolute values exceeds 2. Otherwise z is in the set. Determine whether or not each number belongs to the Mandelbrot set. (3 points)

Problem: a) $z = i$

YES / NO

b) $z = 1 + i$

YES / NO

5. Keep computing until an absolute value goes over 2 (then NO) or the values start repeating (then YES). $|a + bi| = \sqrt{a^2 + b^2}$.

a) $z = i: |i| = 1$

$$z^2 + z = i^2 + i = -1 + i, \quad |-1 + i| = \sqrt{1 + 1} = \sqrt{2} \approx 1.4$$

$$(z^2 + z)^2 + z = (-1 + i)^2 + i = (1 - 2i + i^2) + i = -2i + i = -i, \quad |-i| = 1$$

$$(-i)^2 + i = -1 + i \text{ again: it repeats } (-1 + i, -i, -1 + i, \dots) \text{ and never passes 2} \quad \boxed{\text{YES}}$$

b) $z = 1 + i: |1 + i| = \sqrt{2} \approx 1.4 \leq 2$

$$z^2 + z = (1 + i)^2 + (1 + i) = (1 + 2i + i^2) + 1 + i = 2i + 1 + i = 1 + 3i$$

$$|1 + 3i| = \sqrt{1 + 9} = \sqrt{10} \approx 3.2 > 2 \quad \boxed{\text{NO}}$$

Trap: square the whole number: $(1 + i)^2 = 1 + 2i + i^2 = 2i$, not $1 + i^2$. Stop the moment one value exceeds 2.